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COMPUTING DEVICE AND COMPUTING NODE

Abstract

A computing node includes a housing. The housing includes two side plates that are disposed opposite to each other in a horizontal direction. Each of the two side plates is provided with a plurality of rolling assemblies, and each rolling assembly is capable of rolling in a same rolling plane. The housing includes a bottom plate having a bottom support surface. The bottom support surface and the rolling plane are coplanar, or, in a vertical direction, the bottom support surface is lower than the rolling plane.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of International Application No. PCT/CN2023/117105, filed on Sep. 5, 2023, which claims priority to Chinese Patent Application No. 202211405995.0, filed on Nov. 10, 2022. The disclosures of the aforementioned applications are hereby incorporated by reference in their entireties.

TECHNICAL FIELD
[0002] Embodiments of the present application relates to the field of computing device technology, and in particular to a computing device and a computing node.
BACKGROUND
[0003] A computing device, such as a server, includes a cabinet and a computing node. The computing node is slidably inserted into the cabinet, allowing for convenient mounting and maintenance of the computing node.
[0004] In the related art, drawer slides are arranged on both left and right sides of the computing node, respectively, to achieve sliding assembly of the computing node. However, unlike a general drawer, the computing node has the characteristics of considerable weight and large size along an insertion direction. During long-term use, the drawer slides are prone to damage, leading to issues such as sticking during sliding, poor sliding of the computing node, and difficult to push and pull, which severely affects usability of the computing device.
[0005] Therefore, it is still a technical problem to be solved for those skilled in the art how to provide a solution to better overcome or alleviate the above defects.
SUMMARY
[0006] Embodiments of the present application provide a computing device and a computing node. The computing node is equipped with a rolling assembly, allowing for simple and effortless operation, and having a relatively long service life. In addition, when the computing node is unassembled, the computing mode can be stably placed on an operating table or desk, which is also conducive to protection of the computing node.
[0007] In a first aspect, embodiments of the present application provide a computing node. The computing node may be a server node and the like. The computing node includes a housing. The housing includes two side plates that are disposed opposite to each other in a horizontal direction. Each of the two side plates is provided with a plurality of rolling assemblies, and each of the rolling assemblies is capable of rolling in a same rolling plane. The housing includes a bottom plate having a bottom support surface. The bottom support surface and the rolling plane are coplanar. Or, in a vertical direction, the bottom support surface is lower than the rolling plane.
[0008] In an embodiment, the housing of the computing node is provided with a rolling assembly, and the computing node may be assembled by the rolling assembly to achieve a push-pull operation of the computing node. Compared with a conventional drawer slide solution, the rolling assembly in the solution has a simpler structure and is less prone to damage, offering higher reliability and ensuring normal operation of the computing node within a longer period. In addition, compared with a sliding solution, force that drives the rolling assembly to roll in the solution may be smaller, thereby facilitating pushing and pulling the computing node.
[0009] In addition, in the vertical direction, the bottom support surface may be lower than the

rolling plane, so that when the computing node is placed on the desk or the operating table, the rolling assembly is not in contact with the desk or the operating table, which is conducive to ensuring stable placement of the computing node. Alternatively, the bottom support surface and the rolling plane may also be coplanar. In an embodiment, when the computing node is placed on the desk or the operating table, although the rolling assembly is in contact with the desk or the operating table, the housing may still be stably placed due to static friction between the bottom support surface and the desk or the operating table.

[0010] In an embodiment, the bottom plate has a bottom wall surface, the bottom wall surface includes a bottom support surface and two offset surfaces, the two offset surfaces are located on two horizontal sides of the bottom support surface, the bottom support surface is lower than the offset surfaces, and the rolling plane is lower than the offset surfaces.

[0011] It should be understood that the computing node provided in the embodiments of the present application may be mounted in the cabinet. In order to accommodate mounting of the computing node, a mounting member is provided in the cabinet. The rolling assembly may roll on the mounting member to achieve the mounting or removal of the computing node. The configuration of the offset surfaces allows for avoidance between the housing and the mounting member, so that space inside the cabinet can be better utilized to accommodate the housing, thereby enhancing integration of the device. Moreover, the configuration also facilitates a larger housing size to increase an internal capacity of the housing, so that the housing can accommodate more or larger electronic components. This can improve a computing capability of the computing node.

[0012] In an embodiment, the housing is provided with a groove, the rolling assembly includes a roller, and the roller is mounted in the groove. In this way, the rolling assembly occupies less horizontal space of the computing node, allowing the housing to be made larger in a horizontal direction. Accordingly, the internal capacity of the housing can be increased, and a quantity or size of the electronic components that can be accommodated in the housing can be increased. This, in turn, can improve the computing capability of the computing node.

[0013] The rolling assembly may also include a roller shaft on which the roller may be mounted, and the rolling assembly may be mounted by the roller shaft.

[0014] In an embodiment, the groove has a lower end opening and a side end opening, the side end opening is located on a horizontal outer wall surface of the side plate, and the lower end opening is located on the offset surface. During actual assembly, the rolling assembly may enter the interior of the groove through the side end opening and be mounted on the side plate, allowing for convenient mounting.

[0015] In an embodiment, a horizontal dimension of the roller is less than or equal to that of the groove; and in a horizontal direction, the roller is assembled in the groove. In this way, the roller does not protrude from the horizontal outer wall surface of the side plate in the horizontal direction, and arrangement of the roller does not occupy horizontal space of the housing, which can maximize the internal capacity of the housing, facilitate mounting of more electronic components or larger electronic components, and further improve performance of the computing node.

[0016] In an embodiment, the groove has only a lower end opening, and the lower end opening is located on each of the offset surfaces. In this case, mounting of the rolling assembly does not occupy horizontal space of the housing, which can increase the internal capacity of the housing, facilitate the mounting of more electronic components or larger electronic components, and further improve the performance of the computing node.

[0017] In an embodiment, the housing may have a main cavity and an accommodation space that are isolated from each other. The main cavity may be used for mounting an electronic component in the form of a processor and is filled with a cooling working medium to achieve cooling of the electronic component. The computing node has a plugging direction, and the accommodation space is located on one side of the main cavity in the plugging direction, and is configured with an external connection component. The external connection component may be used to be connected

to the electronic component in the main cavity, or may be in communication with the cooling work medium in the main cavity.
[0018] By adopting the solution, a housing wall plate of the housing can protect the external connection component in the accommodation space, which can reduce damage to the external connection component during transportation and handling, and can also improve structural strength of the entire housing. An appearance design of the housing is also neater.
[0019] In an embodiment, the accommodation space includes a power signal chamber and a high-speed signal chamber that are isolated from each other, the external connection component includes a power cable and a high-speed signal cable, the power cable is disposed in the power signal chamber, and a rear wall of the power signal chamber is provided with a power connector, the high-speed signal cable is disposed in the high-speed signal chamber, and a rear wall of the high-speed signal chamber is provided with a high-speed signal connector. The rear wall of the power signal chamber and the rear wall of the high-speed signal chamber may be an integrated structure to enhance structural strength of the computing node, or the rear wall of the power signal chamber and the rear wall of the high-speed signal chamber may also be a split structure.
[0020] The main cavity is filled with the cooling working medium, and the accommodation space is further provided with a working medium inlet pipe and a working medium outlet pipe for realizing the cooling working medium entering and exiting the main cavity.
[0021] In an embodiment, the housing may have an avoidance portion on two plate bodies that are disposed opposite to each other in a vertical direction. The avoidance portion is used to provide avoidance for the working medium inlet pipe and the working medium outlet pipe. Thus, a size of the housing in the vertical direction can be reduced. The avoidance portion may be a hole-type structure that penetrates a corresponding plate body in the vertical direction, or may be a groove-type structure that does not penetrate the corresponding plate body in the vertical direction.
[0022] The accommodation space may also be provided with an insertion guide assembly for guiding the mounting of the computing node inside the cabinet.
[0023] In an embodiment, the housing includes an upper housing and a lower housing that are mated, the upper housing has a protruding end portion protruding from the lower housing on one end thereof, and the protruding end portion is provided with a protrusion extending toward a side where the lower housing is located. The protrusion may be formed as a handle for pushing, pulling and handling the computing node to facilitate manual operation by the staffs. In addition, the protrusion is arranged to extend toward the side where the lower housing is located, and a size of the computing node in an up-down direction (in the vertical direction) may not be increased, which is conducive to saving mounting space.
[0024] In a second aspect, the embodiments of the present application provide a computing node. The computing node may be a server and the like. The computing device includes a cabinet and at least one computing node. The cabinet has a mounting member. The computing node is a computing node involved in the first aspect or embodiments of the first aspect, which may be in contact with the mounting member through the rolling assembly. Based on the configuration of the rolling assembly, the push-pull operation of the computing node inside the cabinet is made easier and requires less effort, and the computing node has a longer service life and less susceptible to damage, which is conducive to ensuring normal operation of the computing device within a longer period of time.

Description

BRIEF DESCRIPTION OF DRAWINGS
[0025] FIG. **1** is a structural diagram of an embodiment of a data center;
[0026] FIG. **2** is a structural diagram of an embodiment of a cabinet and its internal computing

device;
[0027] FIG. **3** is a structural diagram of an embodiment of a mounting member;
[0028] FIG. **4** is a structural diagram of the mounting member in FIG. **3**;
[0029] FIG. **5** is a structural diagram of a computing node;
[0030] FIG. **6** is a structural diagram of FIG. **5** from another perspective;
[0031] FIG. **7** is a sectional view of FIG. **5** in an A-A direction;
[0032] FIG. **8** is a partial sectional view of another computing node;
[0033] FIG. **9** is a partial sectional view of another computing node;
[0034] FIG. **10** is a structural diagram of a lower housing;
[0035] FIG. **11** is a partial structural diagram of FIG. **10**;
[0036] FIG. **12** is a structural diagram of an upper housing; and
[0037] FIG. **13** is an exploded view of FIG. **12**.
[0038] Reference numerals are as follows: [0039] **100** data center, **101** computer room; [0040] **200** computing device, **201** cabinet, **201***a* door body, **201***b* mounting member, **202** computing node; [0041] **1** housing, **11** upper housing, **111** body, **111***a* first reinforcing rib, **111***b* protruding end portion, **111***b***-1** protrusion, **111***c* second avoidance portion, **112** second reinforcing rib, **12** lower housing, **121** bottom plate, **121***a* bottom support surface, **121***b* offset surface, **121***c* first notch groove, **121***c***-1** first front end wall, **121***d* first avoidance portion, **122** side plate, **122***a* groove, **122***b* second notch groove, **122***b***-1** second front end wall, **123** front plate, **124** rear plate, **125** partition plate, **126** self-locking buckle, **13** sealing assembly, **14** main cavity, **15** accommodation space, **151** power signal chamber, **151***a* power cable, **151***b* power connector, **152** high-speed signal chamber, **152***a* high-speed signal cable, **152***b* high-speed signal connector, **153** working medium inlet pipe, **154** working medium outlet pipe, **155** insertion guide assembly, **156** pressure relief valve, **157** signal connector, **16** sheet metal member; and [0042] **2** rolling assembly, **21** roller shaft, **22** roller, **2***a* rolling plane.

DESCRIPTION OF EMBODIMENTS

[0043] In order to make those skilled in the art better understand the technical solution in the present application, the present application is provided in conjunction with accompanying drawings and embodiments.
[0044] In the embodiments of the present application, the terms "first" and "second" are used only for descriptive purposes, and should not be interpreted as indicating or implying relative importance or implicitly specifying a quantity of the technical features indicated. Thus, features defined by "first" and "second" may explicitly or implicitly include one or more of such features.
[0045] In the embodiments of the present application, "a plurality of" represents two or more; and when "a plurality of" is used to indicate a quantity of several components, it does not represent a relational relationship among these components in terms of quantity.
[0046] In the description of the embodiments of the present application, it should be noted that, unless otherwise specified and limited, terms "mounted", "interconnected" and "connected" are to be understood in a broad sense. For example, "connected" may be detachably connected or non-detachably connected, or may be directly connected or indirectly connected by an intermediate medium.
[0047] In the description of the embodiments of the present application, terms "comprising", "including" or any other variations thereof are intended to cover non-exclusive inclusions, such that a process, method, article, or apparatus that includes a series of elements not only includes those elements, but also includes other elements that are not expressly listed, or further includes elements inherent to the process, method, article, or apparatus. In the absence of additional limitations, elements limited by the sentence "comprising a . . . " does not exclude the presence of additional identical elements in the process, method, article, or apparatus that includes the elements.
[0048] In the embodiments of the present application, "and/or" only refers to an association relationship between association objects, indicating that three relationships may exist. For example,

A and/or B may represent: A exists independently, A and B exist simultaneously, and B exists independently. In addition, the character "/" in the context generally indicates that the association objects are in an "or" relationship.

[0049] With rapid development of communication technology, Internet service providers, enterprises, research institutions, and others generally start to build data centers (also known as computing clusters) that serve as bearers for storage, computing, and transmission functions to meet the demand for data usage. There may be various structure forms for a data center.

[0050] As shown in FIGS. **1** and **2**, FIG. **1** is a structural diagram of an embodiment of a data center, and FIG. **2** is a structural diagram of an embodiment of a cabinet and its internal computing device.

[0051] As shown in FIG. **1**, in some embodiments, the data center **100** may include a computer room **101** and at least one computing device **200**. The computer room **101** is used for providing an isolation environment for the data center **100** to isolate the computing device **200** from the outside world. The computer room **101** may be a permanent house, or a temporary shelter such as a tent or prefab house, or a carrier capable of providing an accommodation space such as a container or cargo box. In some applications, at least one side wall of the computer room **101** may be provided with an entrance/exit door to facilitate staffs and others access to the data center **100**. There is no limitation on a structure model, open/closing mode, or opening/closing control policy of the entrance/exit door, as long as the technical effect that the staffs enter and exit the data center **100** is achieved.

[0052] In some embodiments, the data center **100** is also equipped with an air-cooling circulation system (not shown in the figure) that ventilates and cools the interior of the computer room **101** by configuring at least one air conditioning device or fan device, so as to ensure internal environment and temperature of the computer room **101**.

[0053] In combination with FIG. **2**, the computing device **200** includes a cabinet **201**. A mounting space is formed in the cabinet **201**, and is used for assembling a computing node **202**. The computing device **200** may be a server, and correspondingly, the computing node **202** may be a server node. In some embodiments, there may be a plurality of computing nodes **202**, and each computing node **202** may be located at different positions in the cabinet **201**. For example, each computing node **202** may be distributed layer by layer inside the cabinet **201**. In some embodiments, there may also be one computing node **202**. In this case, the computing node **202** may be referred to as a server.

[0054] The cabinet **201** may be configured with a door body **201***a*. When the door body **201***a* is in an open state, the interior of computing device **200** may be in an exposed state, to facilitate mounting, overhaul, replacement and maintenance of each computing node **202**. A quantity of the door bodies **201***a* may not be limited, which is associated with a structure model and a dimension of the computing node **202**. An opening and closing mode of the door body **201***a* includes, but is not limited to, rotational opening and closing, sliding opening and closing, rolling opening and closing and the like. A lock component matched with the door body **201***a* may refer to related art, and is not limited here.

[0055] The cabinet **201** may be provided with a mounting member for achieving support and assembly of the computing node **202** in the cabinet **201**. There may be various structure forms of the mounting member, as long as the mounting member meets usage requirements, which is not limited in embodiments of the present disclosure.

[0056] As shown in FIGS. **3** and **4**, FIG. **3** is a structural diagram of an embodiment of a mounting member, and FIG. **4** is a structural diagram of the mounting member in FIG. **3**.

[0057] As shown in FIG. **3**, in an example embodiment, a mounting member **201***b* may be an L-shaped plate member, and includes a horizontal plate portion **201***b***-1** and a vertical plate portion **201***b***-2**. The vertical plate portion **201***b***-2** may be mounted on an inner wall of the cabinet **201**. The mounting method includes but is not limited to welding, screw connection, riveting, and snapfitting, as long as mounting reliability can be ensured. The horizontal plate portion **201**_b_**-1** may provide support for the computing node **202**, and space between two horizontal plate portions **201**_b_**-1** may be used to form an air duct for heat dissipation of the computing node **202**.

[0058] It should be understood that in some scenarios, a structure of the computing node **202** may also be designed so that the computing node **202** may occupy the space between the two horizontal portions **201**_b_**-1**. This allows for more efficient use of space inside the cabinet **201**, thereby increasing integration of the device.

[0059] In some other variations, the two horizontal plate portions **201**_b_**-1** may also be an integrated structure, that is, the mounting member **201**_b_ may also generally be U-shaped. In this case, if a heat dissipation requirement exists, a hollow structure may also be disposed on a bottom plate of the mounting member **201**_b_; alternatively, the mounting member **201**_b_ may also directly adopt a plate member and be secured inside the cabinet **201** by means of methods such as welding, insertion or screw connection, which is also feasible.

[0060] A plurality of connection interfaces may be provided inside the cabinet **201**, and are used for connecting the computing node **202**. The connection interfaces include, but are not limited to, power interfaces, high-speed signal structures, cooling working medium interfaces and the like.

[0061] As shown in FIGS. **5** to **7**, FIG. **5** is a structural diagram of a computing node, FIG. **6** is a structural diagram of FIG. **5** from another perspective, and FIG. **7** is a sectional view of FIG. **5** in an A-A direction.

[0062] As shown in FIGS. **5** and **6**, the computing node **202** includes a housing **1** having an internal cavity where an electronic component in the form of a processor is mounted. The housing **1** as a whole may be cuboid-shaped.

[0063] For ease of description, "front", "back", "up", "down", "left", and "right" directions may be defined by taking directions and positional relations in FIG. **4** for example. A front-back direction is a mounting direction of the computing node **202**, where a motion of controlling the computing node **202** to move further into the cabinet **201** is considered as a backward motion, and a motion of controlling the computing node **202** to gradually move out of the cabinet **201** is considered as a forward motion. The computing node **202** may move along the front-back direction in the cabinet **201**. In some embodiments, the front-back direction is also referred to as a length direction or a longitudinal direction. In combination with FIG. **2**, an up-down direction is a direction in which each computing node **202** is stacked in the cabinet **201**. In some embodiments, the up-down direction is also referred to as a height direction, a thickness direction, or a vertical direction. A direction perpendicular to the up-down direction and the front-back direction is a left-right direction. In some embodiments, the left-right direction is also referred to as a width direction or a horizontal direction.

[0064] As shown in FIG. **7**, a plurality of rolling assemblies **2** are disposed on two horizontal sides of the housing **1**. Each of the rolling assemblies **2** is capable of rolling in a same rolling plane **2**_a_. In combination with FIG. **3**, the rolling plane **2**_a_ may be an upper surface of the horizontal plate portion **201**_b_**-1**; the computing node **202** can roll on the mounting member **201**_b_ by the rolling assembly **2**, thereby achieving insertion assembly and pull-out of the computing node **202** inside the cabinet **201**. Compared with a conventional drawer slide solution, the rolling assembly **2** in the solution has a simpler structure and is less prone to damage, offering higher reliability and ensuring normal operation of the computing node **202** within a longer period. In addition, compared with a sliding solution, force that drives the rolling assembly **2** to roll in the solution may be smaller, thereby facilitating pushing and pulling the computing node **202**.

[0065] A quantity of the rolling assemblies **2** disposed on the two horizontal sides of the housing **1** is not specified, and needs to be determined in combination with a size of the computing node **202** in the front-back direction and weight of the computing node. In the embodiment of FIGS. **5** and **6**, eight rolling assemblies **2** are disposed on the two horizontal sides of the housing **1**.

[0066] The housing **1** includes a bottom plate **121** having a bottom support surface **121**_a_. Or, in the up-down direction, the bottom support surface **121***a* may be lower than the rolling plane **2***a*. Thus, when the computing node **202** is placed on a desk or a working table (that is, not assembled in the cabinet **201**), the rolling assemblies **2** are not in contact with the desk or the operating table, which is conducive to ensuring stable placement of the computing node **202**. In an embodiment, the bottom support surface **121***a* and the rolling plane **2***a* may be parallel, or may be arranged with an angle formed therebetween; in the solution where the bottom support surface **121***a* and the rolling plane **2***a* are arranged with an angle formed therebetween, an included angle value between the bottom support surface **121***a* and the rolling plane **2***a* is not limited here.

[0067] It should be understood that, the bottom support surface **121***a* and the rolling plane **2***a* may also be coplanar. In an embodiment, when the computing node **202** is placed on the desk or the operating table, although the rolling assembly **2** is not in contact with the desk or the operating table, the housing **1** may still be stably placed due to static friction between the bottom support surface **121***a* and the desk or the operating table.

[0068] As still shown in FIG. **3**, the housing **1** may further include two side plates **122**, the two side plates **122** may be disposed opposite to each other in a horizontal direction (e.g., a left-right direction), the side plates **122** are located above the bottom plate **121**, and the rolling assembly **2** may be mounted on the side plates **122**. The housing **1** may also be provided with a groove **122***a*. The rolling assembly **2** may include a roller shaft **21** and a roller **22**, the roller shaft **21** may be fixedly mounted on the side plate **122**, the roller **22** may be mounted on the roller shaft **21** by a bearing or other assembly, and may rotate relative to the roller **21**, so as to drive the computing node **202** to move inside the cabinet **201**. It should be understood that the roller shaft **21** may also be fixedly assembled with the roller **22**. In this case, the roller **21** may be mounted on the side plate **122** by a bearing or other assembly, so as to achieve relative rotation of the roller **21** and the side panel **122**, thereby achieving movement of the computing node **202** in the cabinet **201**. In the horizontal direction, the roller **22** may be partially or completely mounted in the groove **122***a*. In this way, the setting of the rolling assembly **2** can occupy relatively less horizontal space of the computing node **202**, and a horizontal dimension of the housing **1** can be made larger. Accordingly, the internal capacity of the housing **1** can also be increased, a quantity or a size of electronic components that can be accommodated in the housing **1** can also be increased, and computing power of the computing node **202** can also be improved.

[0069] The bottom plate **121** may have a bottom wall surface. The bottom wall surface includes a bottom support surface **121***a* and two offset surfaces **121***b*. The two offset surfaces **121***b* may be located on two vertical sides of the bottom support surface **121***a*, the groove **122***a* has a lower end opening, the lower end opening may be located on each of the offset surfaces **121***b*; and the bottom support surface **121***a* may be lower than the offset surfaces **121***b*, and the rolling plane **2***a* may be lower than the offset surfaces **121***b*. In this way, the roller **22** may protrude downward from the offset surfaces **121***b* and smoothly be in contact with the mounting member **201***b*, thereby realizing movement of the computing node **202** in the cabinet **201**.

[0070] Based on an offset configuration of the offset surfaces **121***b* and the bottom support surface **121***a* in the up-down direction, the bottom plate **121** may form first notch grooves **121***c* on two horizontal sides. The configuration, on one hand, can accommodate mounting of the rolling assembly **2** to ensure contact between the rolling assembly **2** and the mounting member **201***b*; and on the other hand, can also reduce material consumption and weight of the housing **1**, thereby facilitating the push-pull operation of the computing node **202**. At the same time, the configuration can also mount the computing node **202** by fully utilizing space between the two horizontal plate portions **201***b***-1**, thereby improving integration of the device.

[0071] In combination with FIG. **6**, the first notch groove **121***c* can extend forward from the rear end of the housing **1** and does not penetrate the entire housing **1** in a front-back direction. In this way, the first notch groove **121***c* can have a first front end wall **121***c***-1**, and projection of the mounting member **201***b* in the front-back direction can fall on the first front end wall **121***c***-1**. In this way, the first front end wall **121**_c_-**1** can be used as a limiting assembly to form a stopper with the mounting member **201**_b_ to limit an insertion depth of the computing node **202** in the cabinet **201**, and can reduce collision force between the computing node **202** and an internal connection interface of the cabinet **201**, which is beneficial to protecting the connection interface, thereby improving a service life of the connection interface. Similarly, the side plate **122** may also be provided with a second notch groove **122**_b_ similar to the first notch groove **121**_c_. The second notch groove **122**_b_ can extend from back to front and does not penetrate the side plate **122**, so that the second notch groove **122**_b_ can have a second front-end wall **122**_b_-**1**. The second front-end wall **122**_b_-**1** can also cooperate with the mounting member **201**_b_, thereby defining the insertion depth of the computing node **202**.

[0072] It should be understood that the first notch groove **121**_c_ and the second notch groove **122**_b_ may be optionally provided; or, the first notch groove **121**_c_ and the second notch groove **122**_b_ may also be through grooves that penetrate along a front-back direction.

[0073] As shown in FIG. **7**, the groove **122**_a_ may extend in a horizontal direction to a horizontal outer wall surface of the side plate **122**. In other words, the groove **122**_a_ may also have a side end opening, and the side end opening may be located on the horizontal outer wall surface of the side plate **122**, so that mounting of the rolling assembly **2** may be relatively easy. The horizontal outer wall surface refers to a wall surface of the side plate **122** away from an internal cavity of the housing **1** in a horizontal direction; in the structure shown in FIG. **7**, the horizontal outer wall surface refers to a right wall surface of the side plate **122**.

[0074] A horizontal dimension of the roller **22** may be less than or equal to that of the groove **122**_a_; and in a horizontal direction, the roller **22** may be assembled in the groove **122**_a_. In this way, the roller **22** does not protrude from the horizontal outer wall surface of the side plate **122**, and the arrangement of the roller **22** does not occupy horizontal space of the housing **1**, which can maximize the internal capacity of the housing **1**, facilitate mounting of more electronic components or larger electronic components, and thus improve performance of the computing node **202**.

[0075] As shown in FIG. **8**, FIG. **8** is a partial sectional view of another computing node.

[0076] As shown in FIG. **8**, the groove **122**_a_ may also be provided in a horizontal middle area of the side plate **122**. In this case, the groove **122**_a_ does not extend to the horizontal outer wall surface of the side plate **122**, and mounting of the roller **22** also does not occupy a horizontal space of the housing **1**. In an embodiment, in order to achieve mounting of the rolling assembly **2**, the side plate **122** or the bottom plate **121** may be configured as a split structure.

[0077] As shown in FIG. **9**, FIG. **9** is a partial sectional view of another computing node.

[0078] In fact, the groove **122**_a_ may also not exist. As shown in FIG. **9**, the roller **22** may be directly mounted on the horizontal outer wall side (that is, one side away from the internal cavity) of the side plate **122**. In this case, the bottom plate **121** may also be not provided with the offset surfaces **121**_b_ and the first notch groove **121**_c_. A structure form of the bottom plate **121** may be relatively simple. Of course, in the solution, the offset surfaces **121**_b_ may also be provided.

[0079] As shown in FIGS. **5** to **9**, the housing **1** may include an upper housing **11** and a lower housing **12**. The upper housing **11** and the lower housing **12** may be mated in an up-down direction and fixedly connected by screw connection, clamping and the like. After being mated, the upper housing **11** and the lower housing **12** may be enclosed to form the internal cavity.

[0080] In order to ensure effective heat dissipation of the electronic component in the housing **1**, a cooling working medium is also filled in the internal cavity of the housing **1**. The cooling working medium may be a liquid phase working medium or a two-phase working medium. In order to avoid leakage of the cooling working medium, a sealing assembly **13** is also configured at a joint of the upper housing **11** and the lower housing **12**. The sealing assembly **13** may be a rubber ring or the like.

[0081] In some embodiments, internal space of the internal cavity may be formed by the lower housing **12**, in which case the upper housing **11** is only equivalent to a cover body. Of course, the internal space of the internal cavity may also be formed by the upper housing **11**, in which case the lower housing **12** is also equivalent to the cover body. In other embodiments, both the upper housing **11** and the lower housing **12** may form cavity space extending in the up-down direction, and after the upper housing **11** and the lower housing **12** are mated, cavity space of the upper housing **11** and cavity space of the lower housing **12** may be in communication with each other to form an internal cavity of the housing **1**. The two implementations may be adopted in specific practice.

[0082] For ease of understanding, the following embodiments of the present application are described by taking the internal space of the internal cavity formed by the lower housing **12** for example. In addition, for ease of description, in the following related descriptions, the internal cavity of the housing **1** is also directly described as the internal cavity of the lower housing **12**.

[0083] As shown in FIGS. **10** and **11**, FIG. **10** is a structural diagram of a lower housing, and FIG. **11** is a partial structural diagram of FIG. **10**.

[0084] As shown in FIG. **10**, the lower housing **12** includes a bottom plate **121**, two side plates **122**, a front plate **123**, and a rear plate **124**. The two side plates **122** may be arranged opposite to each other in a left-right direction, and the front plate **123** and the rear plate **124** may be arranged opposite to each other in a front-back direction. The bottom plate **121**, the two side plates **122**, the front plate **123**, and the rear plate **124** may be integrally formed, and for example, may be integrally cast and formed, so as to simplify a preparation process of the lower housing **12**; of course, these plates may also be independently manufactured and formed, and then connected by screw connection, welding, riveting, clamping and the like. It should be noted that when a connection method such as screw connection, riveting, clamping and the like that cannot directly form a sealed connection is adopted, a sealing member in the form of a sealing ring or the like needs to be arranged between plates to ensure sealing performance of the lower housing **12**.

[0085] The rear plate **124** is not disposed at a rear end of the bottom plate **121**, so that the rear plate **124** may divide space between the bottom plate **121** and the two side plates **122** into two parts, which are respectively a main cavity **14** enclosed by the front plate **123**, the two side plates **122**, the rear plate **124** and the bottom plate **121**, and an accommodation space **15** enclosed by the rear plate **124**, the two side plates **122** and the bottom plate **121**. The main cavity **14** may be located at a front side of the accommodation space **15**, and the upper housing **11** may cover above the main cavity **14** and the accommodation space **15**. The main cavity **14** is configured with an electronic component in the form of a processor and is filled with a cooling working medium used for meeting cooling and heat dissipation requirements of the electronic component; and the accommodation space **15** may be provided with an external connection component used for being connected to a connection interface in the cabinet **201**.

[0086] By adopting the solution, the upper housing **11** and the lower housing **12** can protect the external connection component in the accommodation space **15**, which can reduce damage to the external connection component during transportation and handling, and can also improve structural strength of the entire housing **1**, and the appearance design of the housing **1** is also neater.

[0087] In combination with FIG. **11**, two partitions **125** extending in the front-back direction may also be configured in the accommodation space **15**. The two partitions **125** may be spaced apart in a left-to-right direction to isolate a power signal chamber **151** and a high-speed signal chamber **152** that are isolated from each other in the accommodation space **15**. The power signal chamber **151** is provided with a power cable **151***a*, and the high-speed signal chamber **152** is provided with a high-speed signal cable **152***a*. The power cable **151***a* and the high-speed signal cable **152***a* are both the external connection components.

[0088] In some embodiments, the housing **1** may also be provided with a sheet metal member **16**. The sheet metal member **16** may be connected to the partitions **125** and the side plate **122**, to form rear walls of the power signal chamber **151** and the high-speed signal chamber **152**, thereby enhancing structural strength of the power signal chamber **151** and the high-speed signal chamber

152. It should be understood that the rear walls of the power signal chamber **151** and the high-speed signal chamber **152** may also be independent of each other, that is, two sheet metal members **16** may be provided. The two sheet metal members **16** may respectively form the rear wall of the power signal chamber **151** and the rear wall of the high-speed signal chamber **152**.

[0089] The rear wall of the power signal chamber **151** may be provided with a power connector **151***b*, and the power cable **151***a* may be connected to the power connector **151***b*. The rear wall of the high-speed signal chamber **152** may be provided with a high-speed signal connector **152***b*, and the high-speed signal cable **152***a* may be connected to the high-speed signal connector **152***b*.

[0090] The external connection component may further include a working medium inlet pipe **153** and a working medium outlet pipe **154**. The working medium inlet pipe **153** and the working medium outlet pipe **154** may be mounted on the rear plate **124**, to achieve circulation of the cooling working medium in the main cavity **14**.

[0091] In combination with FIG. **6**, a first avoidance portion **121***d* may also be provided on the bottom plate **121** of the lower housing **12**, and a setting position of the first avoidance portion **121***d* may correspond to a mounting position of the working medium inlet pipe **153** and the working medium outlet pipe **154**, so that the working medium inlet pipe **153** and the working medium outlet pipe **154** may be partially located in the first avoidance portion **121***d*, thereby reducing a size of the housing **1** in the up-down direction. The first avoidance portion **121***d* may be a hole-type structure that penetrates the bottom plate **121** in the up-down direction, or may be a groove-type structure that does not penetrate the bottom plate **121** from top to bottom.

[0092] The accommodation space **15** may also be provided with an insertion guide assembly **155** for guiding an assembly direction of the computing node **202**. A structure of the insertion guide assembly **155** is not limited in the embodiments of the present application. In practical applications, those skilled in the art may configure a structure of the insertion guide assembly **155** based on a structure of a guide matching assembly provided in the cabinet **201**, as long as the technical effect of insertion guidance can be achieved. For example, as shown in FIG. **11**, the insertion guide assembly **155** may be provided with a guide hole **155***a*, and the guide matching assembly inside the cabinet **201** may be a guide column and the like. When the computing node **202** is inserted, the guide column may be inserted into the guide hole **155***a*.

[0093] In some embodiments, a pressure relief valve **156** and a signal connector **157** may also be provided in the accommodation space **15**, and the external connection component includes the pressure relief valve **156** and the signal connector **157**. The pressure relief valve **156** is used to open when pressure in the main cavity **14** is too high, so as to reduce the pressure in the main cavity **14**, thereby improving safety performance, and the signal connector **157** is used for an external signal cable.

[0094] In combination with FIG. **5**, a front end of the lower housing **12** may be provided with a self-locking buckle **126**, and the self-locking buckle **126** can cooperate with the cabinet **201** to achieve mounting and fixation of the computing node **202** inside the cabinet **201**. A structure of the self-locking buckle **126** is not limited here, and in practical applications, may be determined by those skilled in the art based on the related art. It shall be understood that the self-locking buckle **126** may also be provided on the upper housing **11**, as long as the self-locking function can be achieved.

[0095] As shown in FIGS. **12** and **13**, FIG. **12** is a structural diagram of an upper housing, and FIG. **13** is a partial structural diagram of FIG. **12**.

[0096] As shown in FIGS. **12** and **13**, the upper housing **11** includes a body **111**, which is basically a plate-like structure, and a plurality of first reinforcing ribs **111***a* are provided on one side toward the main cavity **14** to improve structural strength of the body **111**. The body **111** may generally be made of aluminum alloy or the like.

[0097] In some embodiments, the upper housing **11** may also include a second reinforcing rib **112**. The second reinforcing rib **112** may be made of a material with higher structural strength such as steel. The second reinforcing rib **112** may be mounted and fixed to the body **111** by a connector in the form of a screw and the like, so as to further improve structural strength of the upper housing **11**.

[0098] In combination with FIGS. **4** and **6**, the front end of the upper housing **11** has a protruding end portion **111***b* protruding from the lower housing **12**, and the protruding end **111***b* may be configured with a protrusion **111***b***-1** extending downward. Since the protrusion **111***b***-1** is arranged to extend downward, a size of the computing node **202** in the up-down direction may not be increased, which is conducive to saving mounting space. At the same time, the protrusion **111***b***-1** may be used as a handle for pushing, pulling and handling the computing node **202** to facilitate manual operation by the staffs.

[0099] In some embodiments, a second avoidance portion **111***c* may also be provided on the upper housing **11**. A setting position of the second avoidance portion **111***c* may correspond to a mounting position of the working medium inlet pipe **153** and the working medium outlet pipe **154**, so that the working medium inlet pipe **153** and the working medium outlet pipe **154** may be partially located in the second avoidance portion **111***c*, thereby reducing the size of the housing **1** in an up-down direction. The second avoidance portion **111***c* may be a hole-type structure that penetrates the upper housing **11** in the up-down direction, or may be a groove-type structure that does not penetrate the upper housing **11** from bottom to top.

[0100] The above are only embodiments of the present application. It should be noted that for those skilled in the art, a plurality of improvements and modifications can be made without departing from the principles of the present application. These improvements and modifications should also be considered to be within the protection scope of the present application.

Claims

1. A computing node, comprising: a housing, wherein the housing comprises: two side plates disposed opposite to each other in a horizontal direction, wherein each of the two side plates is provided with a plurality of rolling assemblies, and each of the rolling assemblies is capable of rolling in a rolling plane; and a bottom plate having a bottom support surface, wherein the bottom support surface and the rolling plane are coplanar or in a vertical direction, and the bottom support surface is lower than the rolling plane.

2. The computing node according to claim 1, wherein the bottom plate comprises a bottom wall surface, the bottom wall surface comprises the bottom support surface and two offset surfaces, the two offset surfaces are located on two horizontal sides of the bottom support surface, the bottom support surface is lower than the offset surfaces, and the rolling plane is lower than the offset surfaces.

3. The computing node according to claim 2, wherein the housing comprises a groove, the rolling assembly comprises a roller, and the roller is mounted in the groove.

4. The computing node according to claim 3, wherein the groove comprises: a lower end opening disposed on the offset surface; and a side end opening disposed on a horizontal outer wall surface of the side plate.

5. The computing node according to claim 4, wherein a horizontal dimension of the roller is less than or equal to that of the groove; and in a horizontal direction, the roller is assembled in the groove.

6. The computing node according to claim 3, wherein the groove comprises a lower end opening disposed on the offset surface.

7. The computing node according to claim 1, wherein the housing further comprises: a main cavity; and an accommodation space isolated from the main cavity, disposed on one side of the main cavity in a plugging direction of the computing node, and is configured with an external connection component.

8. The computing node according to claim 7, wherein the accommodation space comprises a power signal chamber and a high-speed signal chamber that are isolated from each other, the external connection component comprises a power cable and a high-speed signal cable, the power cable is disposed in the power signal chamber, and a rear wall of the power signal chamber is provided with a power connector, the high-speed signal cable is disposed in the high-speed signal chamber, and a rear wall of the high-speed signal chamber is provided with a high-speed signal connector.
9. The computing node according to claim 7, wherein the main cavity is filled with a cooling working medium, and the accommodation space is further provided with a working medium inlet pipe and a working medium outlet pipe.
10. The computing node according to claim 7, wherein the accommodation space is further provided with an insertion guide assembly.
11. The computing node according to claim 1, wherein the housing further comprises an upper housing and a lower housing that are mated, the upper housing has a protruding end portion that protrudes from the lower housing on one end thereof, and the protruding end portion is provided with a protrusion that extends toward a side where the lower housing is located.
12. The computing node according to claim 1, wherein the housing comprises an avoidance portion on two plate bodies that are disposed opposite to each other in the vertical direction, the avoidance portion provides avoidance for a working medium inlet pipe and a working medium outlet pipe.
13. A computing device, comprising: a cabinet having a mounting member; and a computing node, comprising: a housing, wherein the housing comprises: two side plates disposed opposite to each other in a horizontal direction, wherein each of the two side plates is provided with a plurality of rolling assemblies, and each of the rolling assemblies is capable of rolling in a rolling plane; and a bottom plate having a bottom support surface, wherein the bottom support surface and the rolling plane are coplanar or in a vertical direction, and the bottom support surface is lower than the rolling plane, and wherein at least one of the rolling assemblies is in contact with the mounting member.
14. The computing device according to claim 13, wherein the bottom plate comprises a bottom wall surface, the bottom wall surface comprises the bottom support surface and two offset surfaces, the two offset surfaces are located on two horizontal sides of the bottom support surface, the bottom support surface is lower than the offset surfaces, and the rolling plane is lower than the offset surfaces.
15. The computing device according to claim 14, wherein the housing comprises a groove, the rolling assembly comprises a roller, and the roller is mounted in the groove.
16. The computing device according to claim 15, wherein the groove comprises: a lower end opening disposed on the offset surface; and a side end opening disposed on a horizontal outer wall surface of the side plate.
17. The computing device according to claim 16, wherein a horizontal dimension of the roller is less than or equal to that of the groove; and in a horizontal direction, the roller is assembled in the groove.
18. The computing device according to claim 15, wherein the groove comprises a lower end opening disposed on the offset surface.
19. The computing device according to claim 13, wherein the housing further comprises: a main cavity; and an accommodation space isolated from the main cavity, disposed on one side of the main cavity in a plugging direction of the computing node, and is configured with an external connection component.
20. The computing device according to claim 19, wherein the accommodation space comprises a power signal chamber and a high-speed signal chamber that are isolated from each other, the external connection component comprises a power cable and a high-speed signal cable, the power cable is disposed in the power signal chamber, and a rear wall of the power signal chamber is provided with a power connector, the high-speed signal cable is disposed in the high-speed signal

chamber, and a rear wall of the high-speed signal chamber is provided with a high-speed signal connector.